Video-Disjoint Semantic Segmentation of Marine Debris

TrashCan-Material — U-Net vs. SegFormer-B0

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Problem and research questions

Task

Given an underwater RGB image, assign each valid pixel one of 17 semantic labels: background, ROV, plant, six animal classes, or eight trash classes. Instance identity is not predicted; ignore label 255 is excluded.

Why it is difficult

Small and thin debris, low contrast, occlusion, variable illumination, and strong class imbalance make foreground recovery difficult.

Research questions

1. On a video-disjoint split, how do scratch U-Net and pretrained SegFormer-B0 compare on foreground and trash overlap?
2. Does mild RGB color jitter improve SegFormer trash overlap while preserving foreground quality?
3. Which errors persist, and what are the measured inference costs?

Metrics: macro foreground mIoU/Dice; trash mIoU averages eight trash classes.

Dataset overview: TrashCan 1.0 Material

Source and data format

- JAMSTEC underwater ROV imagery distributed as the TrashCan 1.0 Material dataset through the University of Minnesota data repository.
- 7,212 sampled RGB JPEG frames and 12,336 polygon annotations in COCO-style JSON files.
- 312 filename-derived video groups; frames from a video are kept together during splitting.
- Annotation JSONs list Attribution-NonCommercial-ShareAlike; check the dataset record for reuse terms.

Semantic label set

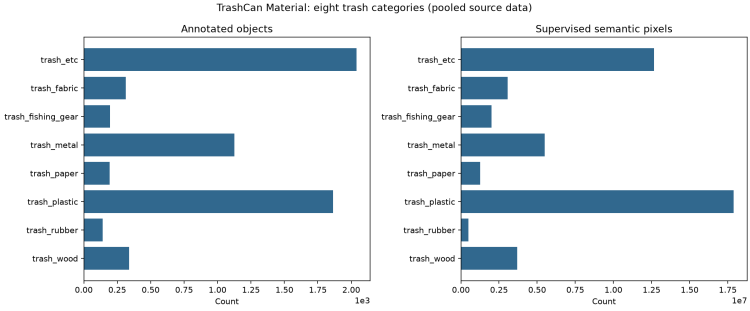
17 evaluated labels: background, ROV, plant, six animal classes, and eight trash classes. Instance identity is not a prediction target; ambiguous cross-class overlaps and padded pixels use ignore ID 255.

Native frame sizes

Resolution	Images	Share
480×360	3,245	45.0%
480×270	3,967	55.0%

Source: Dataset counts and dimensions from the project EDA and fixed manifest.

Detailed EDA: class and pixel imbalance



Background dominance

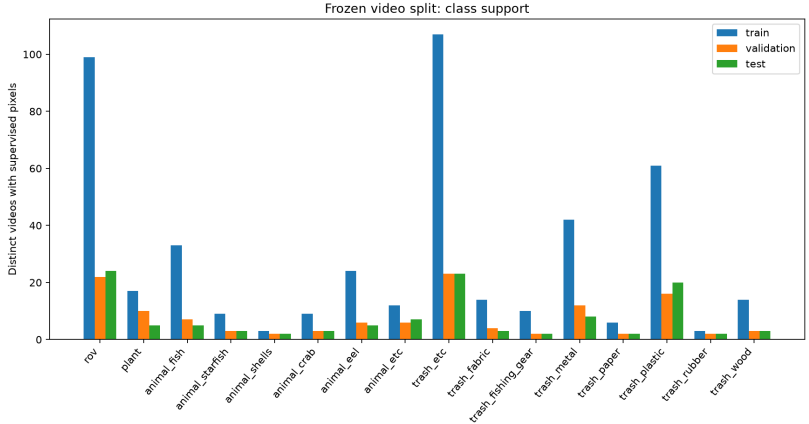
Background accounts for 90.33% of valid test pixels. Pixel accuracy can therefore look strong while debris is missed.

Uneven trash support

Trash annotations range from 142 rubber instances to 2,039 trash_etc instances. Native supervised pixels range from 0.493M rubber to 17.88M plastic.

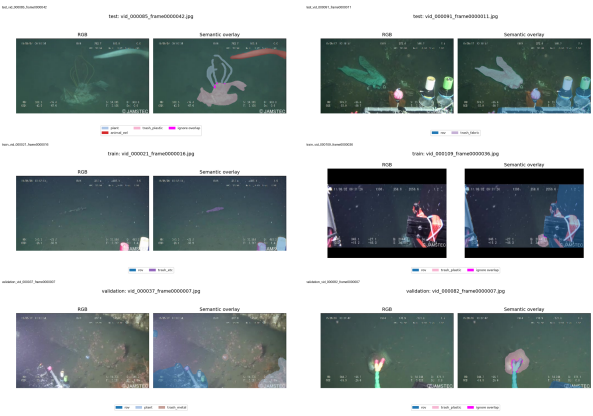
Left: pooled annotation-instance counts and rasterized semantic pixels for the eight trash labels.

Detailed EDA: class support across video splits



Seed 36: train 5,057 images / 207 videos; validation 1,077 / 54; test 1,078 / 51. Zero video-ID overlap; original train/validation shared 127 IDs.

Detailed EDA: inspect the polygon annotations



Visual checks

Representative train, validation, and test examples show the raw RGB frame alongside its polygon-derived semantic overlay.

What to watch

Thin or low-contrast objects can be hard to delineate. Cross-class polygon overlaps are marked ignore rather than arbitrarily assigned; rare classes also have limited video support.

Preprocessing: from frames to a training batch

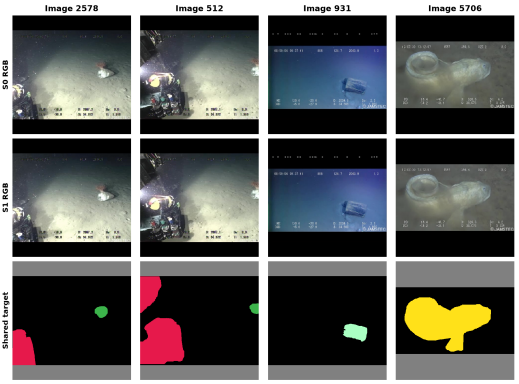
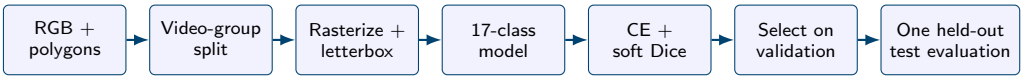


Image and mask pipeline

1. Rasterize polygons at native size; same-class regions are unioned and cross-class overlap pixels become ignore 255.
2. Letterbox RGB and masks to 512×512: bilinear image resize, nearest-neighbor mask resize, black image padding, ignore-label mask padding.
3. Normalize RGB with ImageNet mean and standard deviation. No geometric augmentation.
4. S1 only: color jitter with $p = 0.5$, brightness/contrast/saturation $[0.9, 1.1]$, hue $[-0.01, 0.01]$; masks stay fixed.

Seed-69 first shuffled batch: IDs 2578, 512, 931, 5706; RGB $4 \times 3 \times 512^2$, masks 4×512^2; S0/S1 masks match.

Shared training and evaluation pipeline



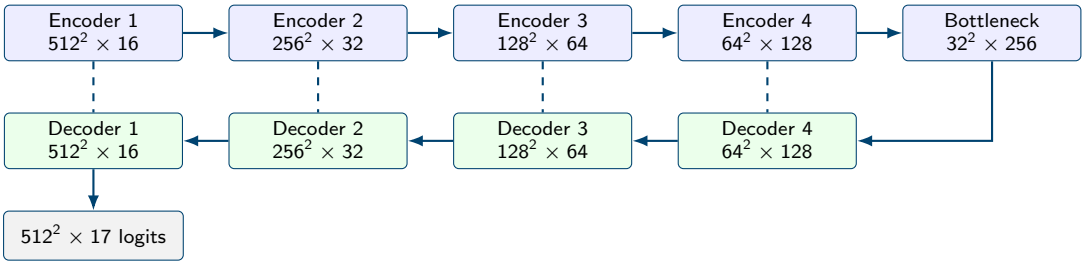
Scoring

Confusion matrix accumulated over valid pixels; class IoU and Dice derived from TP, FP, FN. Foreground metrics macro-average classes 1–16; trash mIoU macro-averages classes 9–16.

Test-set separation

Validation selects the checkpoint. The fixed test split is evaluated only after training; test results do not choose epochs or settings.

Model architecture: scratch U-Net



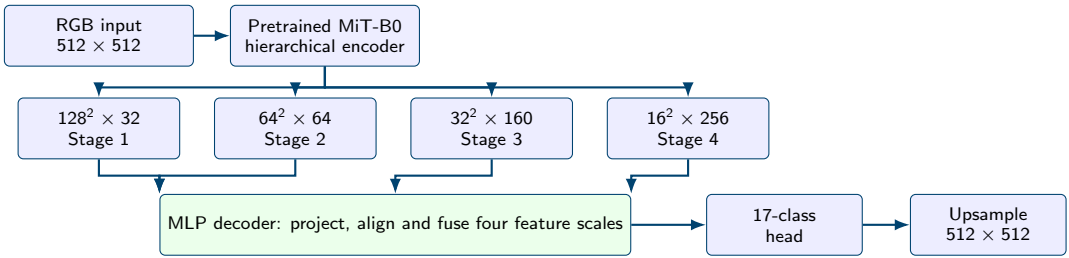
Design

Four encoder/decoder scales; double 3×3 convolutions with GroupNorm and ReLU. Max pooling downsamples; bilinear upsampling and skip concatenations recover spatial detail.

Role and measured size

Scratch-trained baseline with a 1×1 17-class output head; 1,964,369 trainable parameters. Compact, but no pretrained feature initialization.

Model architecture: pretrained SegFormer-B0



Encoder and decoder

ADE20K-pretrained MiT-B0 produces four feature maps at 1/4 to 1/32 input resolution. An MLP decoder aligns and fuses the scales before the new 17-class head.

Fine-tuning and variants

All encoder, decoder, and head parameters are trainable: 3,718,513 total. S0 uses standard RGB; S1 adds mild RGB color jitter during training. Checkpoint revision: 489d5cd.

Loss and training setup

Objective

$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \mathcal{L}_{\mathrm{softDice,FG}}$ (weight 1 each). CE ignores label 255; soft Dice averages foreground classes present in the batch. All-ignore batches are skipped.

Why combine them?

CE supervises pixel-wise labels; foreground Dice counters dominance by large background regions.

Optimization and budget

AdamW, weight decay 0.01, batch size 4, up to 40 epochs, early-stopping patience 8. FP32 with no scheduler or mixed precision. Learning rates: U-Net 10^{-3}; SegFormer 6×10^{-5}. Seeds: 69420, 67, 69.

Checkpoint rule

Select best validation foreground mIoU; ties use lower validation loss, then earlier epoch. Each epoch checkpoint records model, optimizer, RNG, history, and stopping state.

Execution environment and run provenance

Hardware and numerical stack

GPU	Tesla T4, 14.56 GiB
Python	3.13.16
PyTorch	2.11.0+cu130
torchvision	0.26.0+cu130
CUDA	13.0
NumPy / Pillow	2.1.3 / 12.3.0
Transformers	4.57.6

Run provenance

Runs were executed across Kaggle and Colab sessions. Seed 67/S0 resumed on Colab from its Kaggle checkpoint after epoch 7; it is one continued run, not an additional seed.

Reproducibility note

Mixed execution profiles can affect runtime and numerical reproducibility. Environment and checkpoint identities are recorded with the bundled run evidence.

Experimental design: three runs per configuration

U vs. SegFormer

Question: does the pretrained SegFormer configuration improve foreground and trash overlap over a compact scratch baseline?

Changed: architecture, initialization, parameter count, and learning rate.

Fixed: split, input, loss, optimizer family, batch, epoch budget, and evaluation protocol.

S0 vs. S1 controlled color ablation

Hypothesis: mild training color jitter improves trash mIoU while preserving foreground quality.

Changed: RGB jitter only; per-seed initial weights and nominal training settings matched.

Predeclared validation rule: trash mIoU gain ≥ 1 pp, with FG mIoU and Dice losses each ≤ 1 pp.

Runs span Kaggle and Colab. Seed 67/S0 resumed on Colab from a Kaggle checkpoint after 7 epochs; this is one continued run, not a new seed. Platform variation and three repeats limit inference.

Held-out results: SegFormer configurations lead

Configuration	FG mIoU (%)	FG Dice (%)	Trash mIoU (%)
U: scratch U-Net	4.52 ± 0.83	7.25 ± 1.21	1.79 ± 0.29
S0: SegFormer, no jitter	15.86 ± 2.47	22.81 ± 2.74	6.60 ± 1.49
S1: SegFormer + jitter	15.41 ± 1.94	22.64 ± 2.23	5.88 ± 0.25

Consistent direction vs. U-Net

At every measured seed, both SegFormer configurations exceed U-Net on test foreground mIoU and trash mIoU.

Not a universal augmentation win

S0 has the highest three-seed average for all three aggregate metrics. Seed-level results show the color effect changes direction; do not interpret these three repeats as a stable general effect.

Source: Mean $\pm$ sample SD over three seeds; same 1,078 held-out images for all nine runs.

Per-seed held-out results

Seed	Run	Best epoch	FG mIoU (%)	FG Dice (%)	Trash mIoU (%)
69420	U	15	3.563	5.852	1.465
69420	S0	15	13.588	20.566	6.403
69420	S1	14	13.964	20.895	6.078
67	U	15	4.901	7.911	1.915
67	S0	20	18.492	25.868	8.172
67	S1	9	17.621	25.158	5.596
69	U	12	5.095	7.986	2.002
69	S0	8	15.489	21.993	5.217
69	S1	22	14.656	21.873	5.964

Consistent direction, modest evidence base

Both SegFormer runs beat U-Net on foreground and trash mIoU at every seed on the same fixd test split.

Best epoch is selected only by validation foreground mIoU (ties: validation loss, then earlier epoch).

Color-jitter ablation: seed-dependent outcome

Validation deltas: S1 − S0 (pp)

Seed	FG mIoU	FG Dice	Trash mIoU	Rule
69420	−1.953	−2.175	−2.425	No
67	−2.131	−2.515	+0.821	No
69	+1.157	+1.917	+1.971	Yes

Predeclared decision

Only seed 69 reaches the required +1 pp validation trash mIoU gain while limiting both foreground losses to 1 pp.

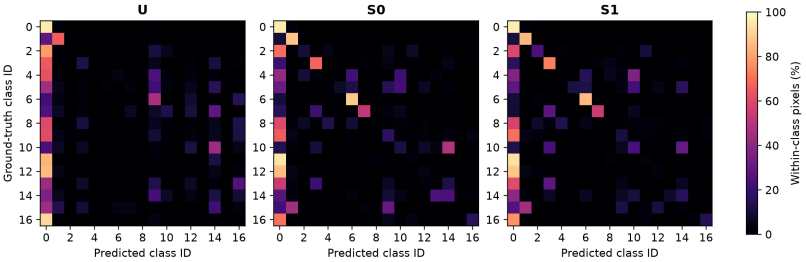
Held-out deltas: S1 − S0 (pp)

Seed	FG mIoU	FG Dice	Trash mIoU	Validation rule
69420	+0.376	+0.330	−0.326	Not met
67	−0.870	−0.709	−2.576	Not met
69	−0.833	−0.120	+0.747	Met

Interpretation

Test trash mIoU improves only at seed 69. S0 has the higher three-seed mean; the evidence does not show a consistent augmentation benefit.

Where the models confuse labels



True trash → background

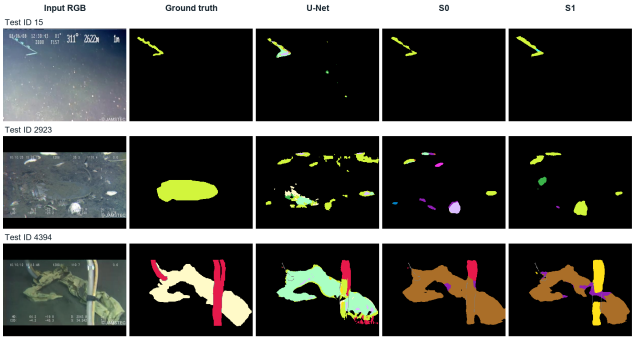
Mean over seeds:
U 56.22%
S0 52.76%
S1 59.83%

Class-specific trade-off

S1–S0 mean IoU: fabric +4.26 pp; plastic −5.87 pp.

Rows are ground truth, columns predictions; each row normalized, then averaged across seeds. Macro trash mIoU weights each trash class equally, unlike pixel-weighted background error.

Qualitative error analysis: matched test cases



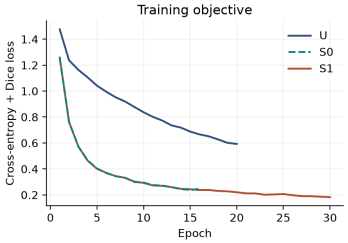
Thin object: on image 15, correctly labeled trash_etc pixels: S1 1,299/1,527; S0 1,095; U 664.

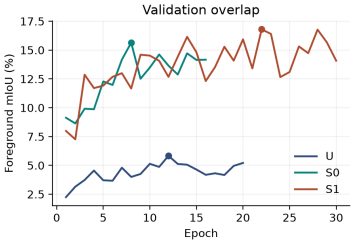
Foreground omission: on image 2923, both SegFormer models send $> 98\%$ of annotated foreground to background.

Material confusion: on image 4394, both SegFormer models localize plastic but label most pixels as paper.

Source: Aligned seed-69 RGB, ground truth, and model predictions; vertical letterbox padding cropped for display.

Learning dynamics and training compute trade-off





Selected / final epoch

U: 12 / 20; S0: 8 / 16; S1: 22 / 30.

Recorded runtime (hours)

Run	Train	Valid.
U	5.69	0.42
S0	4.51	0.29
S1	8.82	0.54

Training-cost interpretation

S1 has the longest recorded training time (8.82 h); S0 took 4.51 h versus U-Net at 5.69 h. Stop epoch and platform variation affect wall-clock totals. Validation overlap fluctuates, so the best validation foreground mIoU selects the checkpoint.

Runtime is wall-clock evidence from mixed Kaggle/Colab sessions, not a controlled cross-model throughput benchmark.

Inference cost: matched forward benchmark

Configuration	Parameters	Batch-4 forward	Amortized / image
U-Net	1.96 M	84.98 ms	21.24 ms
SegFormer S0	3.72 M	110.95 ms	27.74 ms
SegFormer S1	3.72 M	111.09 ms	27.77 ms

Benchmark protocol

Tesla T4, FP32, preloaded batch of four; 10 warm-up iterations followed by 50 measured forwards. Models were already on the accelerator.

How to read the times

Forward-only: excludes image loading, host-to-device transfer, and postprocessing.
Per-image values divide batch median by four; they are not batch-one latency.

U-Net uses fewer parameters and has the lowest measured forward time; both SegFormer variants have effectively identical costs.

Discussion

Model comparison

- On this fixed test split, both SegFormer configurations outperform scratch U-Net for foreground and trash mIoU at each seed.
- U-Net is smaller and faster per image in the matched forward benchmark; SegFormer yields better measured overlap at higher compute/parameter cost.

Color-jitter ablation

- No consistent test-set gain: S0 has the higher three-seed mean across all three aggregate metrics.
- The test trash-mIoU change varies by seed, from -2.576 pp to $+0.747$ pp.

Interpretation

The U-Net/SegFormer comparison changes architecture, initialization, parameter count, and learning rate together, so it does not isolate architecture as a causal effect.

Limitations

Scope of the evidence

- Results use one fixed video-ID split; distinct IDs may still share scenes or dive locations.
- Three seeds and mixed Kaggle/Colab execution profiles limit uncertainty estimates.
- The three seed runs evaluate the same held-out images, not independent test datasets.
- Rare classes, label ambiguity, thin objects, background omissions, and material confusion remain difficult.

Conclusion

Takeaway

On this video-disjoint TrashCan split, both SegFormer configurations outperform the scratch U-Net for foreground and trash overlap at each seed. The pretrained models improve measured overlap at higher parameter and inference cost; the color-jitter benefit is not consistent across seeds.

Next experiments

Run more seeds on one platform, evaluate an independent-dive test set, and analyze rare classes and object boundaries; select any test-time augmentation using validation data only.

Reproducibility and references

Evidence bundled with the project

- `evidence/manifest.json`: split membership and rasterization policy.
- `evidence/runs/`: effective configs, histories, test metrics, and confusion matrices for nine runs.
- `verified_results.json`: audited result summary and run/checkpoint identities.
- `REPRODUCIBILITY.md`: environment, evidence layout, and verification guidance.

Selected references

Hong, Fulton & Sattar. "TrashCan: A Semantically-Segmented Dataset towards Visual Detection of Marine Debris." 2020.

Ronneberger, Fischer & Brox. "U-Net: Convolutional Networks for Biomedical Image Segmentation." 2015.

Xie et al. "SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers." 2021.

Pretrained SegFormer-B0 ADE20K checkpoint and config revision are pinned in the run configs.

AI assistance was used during project development and presentation drafting; the group remains responsible for verification and assignment-specific disclosure.

Thank you

Questions?

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